

Local Evidence Is Not Completion: Evidence-Condition Geometry for Bounded Completion-Audit Workflows

Anonymous Submission

Abstract

Bounded completion-audit workflows with declared source-route scopes often need to decide whether a task is complete before the remaining workload is known. Common stop signals, including no-new findings, agent agreement, stable summaries, and self-reported completion, can be useful local evidence, but they do not specify the conditions under which that evidence was produced. The key question is not whether a workflow has stopped, but whether the evidence condition under which it stopped can support the declared scope of the completion claim. We study this certificate mismatch and introduce source-route evidence-condition geometry as a runtime diagnostic. A source-route stratum records both where the workflow searched and which audit route it applied. We then define an evidence-condition controller that rejects SAFE when the current condition is too localized, repairs weak plausible strata, and returns SAFE, CONTINUE, or ABSTAIN. Across four bounded completion-audit tests, homogeneous route reuse produces localized evidence conditions and would lead to false certification if accepted as a scope-wide stop. Broader source-route exposure improves completion eligibility, while an external `urllib3` audit shows the key boundary: broad exposure is not sufficient for SAFE when residual evidence remains. The paper provides a diagnostic and control principle for bounded audit completion, not a universal completion guarantee for open-world agent workflows.

1 Introduction

This paper evaluates bounded completion-audit workflows with declared source-route scopes. These are workflows whose intended evidence region is bounded, such as a repository snapshot or document set, and whose completion claim is stated over a declared set of sources and audit routes. Examples include searching a repository snapshot for all instances of a behavior, auditing a document set for policy exceptions, checking whether an implementation covers declared failure-mode routes, or deciding whether a multi-agent audit has exhausted useful evidence within its declared scope. Such workflows must eventually stop. The difficulty is that the absence of newly found evidence is not the same object as evidence that the intended bounded workload is complete.

The failure studied in this paper is *certificate mismatch*. A workflow may produce a plausible local stop signal: repeated scans return no new findings, agents agree with each other,

subtasks appear closed, or a summary becomes stable. These signals are conditioned on the parts of the task that were searched and on the routes by which they were searched. If the workflow repeatedly applies the same route to the same subset of sources, a no-new signal may certify only local exhaustion. It does not automatically support a global claim that the task is complete over its intended scope.

Consider a code-audit workflow. Auditing a file for timeout behavior does not certify that the same file has been audited for TLS behavior, exception behavior, retry behavior, or cleanup behavior. The source may be familiar, the agents may agree, and the workflow may have stopped finding timeout-related evidence; none of that establishes that other routes over the same source have been exhausted. The stop evidence is local in its evidence condition, while the completion claim is global in scope.

We formalize this observation with *evidence-condition geometry*. The basic unit is a source-route stratum, consisting of a bounded evidence region and an audit or search route. At runtime, the workflow accumulates exposure counts over these strata. The normalized exposure distribution describes the condition under which completion evidence was produced. When this distribution is localized, stop evidence is locally conditioned. It may be useful, but it is not a global completion certificate.

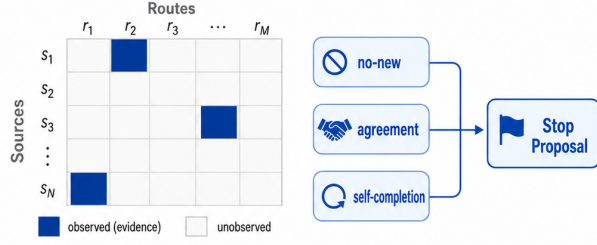
Based on this geometry, we define an evidence-condition controller. At a proposed stop, the controller computes the source-route exposure condition, checks whether the condition is broad enough to make the completion claim eligible for certification, and triggers repair over weak plausible strata when necessary. The controller is intentionally conservative: broad exposure gives completion eligibility, not automatic safety. A SAFE decision additionally requires that repair or audit reveal no residual evidence. If residual evidence appears, the controller outputs CONTINUE; if budget is exhausted without a defensible certificate, it outputs ABSTAIN.

Our experiments evaluate this controller on four bounded completion-audit task families: two generated scored tasks and two external real-repository audits with pattern-defined oracles. These tasks are not presented as a benchmark for the best item discovery policy. Instead, they test whether source-route exposure localization diagnoses unsafe stop claims and whether a controller can avoid accepting locally conditioned evidence as global completion proof.

This paper makes three contributions. First, it formulates false completion as a certificate mismatch between the declared scope of a stop claim and the source-route condition

Local Evidence Is Not a Global Completion Certificate

1 Observed Evidence Condition



2 unsafe promotion

Promoting local stop evidence to a global claim is invalid.

3 Global Completion Claim

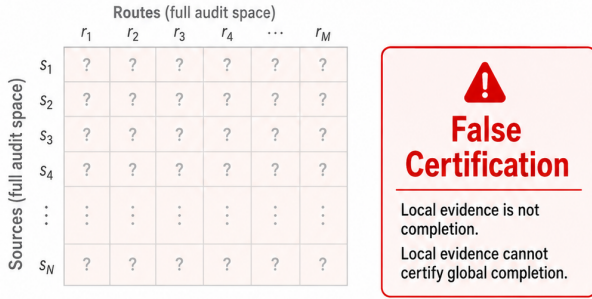


Figure 1: Certificate mismatch from local stop evidence. A workflow may observe no-new findings or agreement under localized source-route exposure. Such evidence can support local exhaustion, but it does not by itself certify completion over the intended source-route space. The evidence-condition controller audits this condition before accepting SAFE.

under which supporting evidence was produced. Second, it introduces source-route exposure geometry and an evidence-condition controller that separates completion eligibility from completion proof, repairing weak plausible strata before accepting SAFE. Third, it provides bounded audit evaluation, including generated audits and external repository audits, showing that homogeneous route reuse creates false certification risk while broader source-route evidence improves eligibility without being sufficient for safety.

We deliberately do not claim that residual-potential is an optimal repair method. It is one mechanism-aligned repair instance used to test whether weak source-route regions can expose residual evidence. The main claim is about the conditions under which completion evidence can support a stop decision.

2 Related Work

High-recall review, active search, and missing mass. Technology-assisted review and total-recall systems study how to retrieve nearly all relevant items under review bud-

gets [Grossman and Cormack, 2011, Cormack and Grossman, 2014, 2015, 2016, Grossman et al., 2016, Roegiest et al., 2015], while active search and active learning choose actions or examples to expose positives efficiently [Lewis and Gale, 1994, Settles, 2009, Garnett et al., 2012, Attenberg et al., 2015]. Missing-mass and unseen-species estimators ask how much probability mass remains unseen [Good, 1953, Chao, 1984, Chao et al., 2014]. Our scored subclasses use recall only for post-hoc false-certification measurement. The controller does not observe oracle totals, recall, or hidden missing mass; it observes the source-route condition under which a stop claim was produced and whether repair reveals residual evidence.

Agent workflows and verification. Tool-using, browser-assisted, multi-agent, and memory-based language agents allocate work across tools, agents, searches, and intermediate states [Yao et al., 2023, Schick et al., 2023, Nakano et al., 2021, Qin et al., 2024, Wu et al., 2023, Park et al., 2023, Shinn et al., 2024, Madaan et al., 2023, Wang et al., 2024]. Agent benchmarks and factual verification benchmarks evaluate task success, tool use, factuality, or evidence support [Yao et al., 2022, Deng et al., 2024, Zhou et al., 2024, Liu et al., 2024, Jimenez et al., 2024, Hendrycks et al., 2021a, Liang et al., 2022, Srivastava et al., 2023, Lin et al., 2022, Thorne et al., 2018, Gao et al., 2023, Min et al., 2023, Schuster et al., 2021]. Our question is narrower and precedes final answer scoring: whether the evidence condition under which a workflow stopped is broad enough for the scope of its completion claim.

Agreement, audit agents, and oversight. Ensembles, self-consistency, verifier-guided systems, and audit agents can improve reliability [Dietterich, 2000, Cobbe et al., 2021, Wang et al., 2023]. Agreement is not useless, but it is unsafe as an unconditioned completion certificate: agents may agree because they repeatedly examined the same local region. Safety and oversight work studies broader failure modes and scalable supervision [Amodei et al., 2016, Christiano et al., 2017, Irving et al., 2018, Leike et al., 2018, Hendrycks et al., 2021b]. This paper targets a specific operational failure within that broader landscape: a workflow may present a global completion certificate whose supporting evidence was produced under a local source-route condition.

3 Problem Setting

We study bounded completion-audit workflows with unknown residual workload and declared source-route scope. A workflow receives a task, allocates work across agents or tools, gathers evidence under a finite budget, and must decide whether the task is complete within that declared scope. Let

$$W = (T, S, R, A, B),$$

where T is the task, S is a set of sources, R is a set of routes, A is the set of agents or tools, and B is a finite budget. A

source is a bounded evidence region, such as a file, module, document, source family, or repository component. A route is an audit or search lens applied to a source.

At runtime, the workflow produces evidence events

$$e_t = (a_t, x_t, r_t, u_t, o_t, c_t),$$

where a_t is the agent or tool, $x_t \in S$ is the source, $r_t \in R$ is the route, u_t is the action, o_t is the observation, and c_t is the cost. Every evidence event is produced under a source-route condition. A source-route stratum is $s = (x, r)$, and the intended evidence-condition space is

$$\Omega = S \times R.$$

The source-route space does not require the workflow to know all true missing items in advance. It requires the intended audit scope to be defined: which sources and routes are relevant to the completion claim. This assumption is weaker than knowing the answer set and stronger than an unbounded open-world search. It matches many practical audits: a system may know which repositories, documents, modules, or evidence sources are in scope, and it may know which audit routes are relevant, while still not knowing how many true findings remain. The controller therefore evaluates the condition under which evidence was gathered, not the unknown total workload itself.

A stop claim is a workflow assertion

C_t : T is complete over the intended workload scope.

A false certification occurs when the workflow accepts SAFE for a stop claim that is not supported by the evaluation oracle:

$$\text{SAFE}(C_t) = \text{true}, \quad \text{complete}(T) = \text{false}.$$

In item-discovery subclasses, this is operationalized as accepting SAFE while bounded-oracle recall is below the completion threshold. In broader completion audits, it means accepting a global completion claim while relevant support, contradiction, or unresolved evidence remains outside the evidence condition. The oracle is used only for evaluation. It is not available to route assignment, repair scoring, eligibility checks, or stop decisions.

4 Evidence-Condition Geometry

Let $v_t(s)$ be the runtime-visible exposure count for source-route stratum s up to time t . Exposure may count visits, scans, route-specific audits, tool calls, or other logged search actions. Define

$$p_{\text{exp}}(t, s) = \frac{v_t(s)}{\sum_{s' \in \Omega} v_t(s')}.$$

This distribution is a point on the source-route coverage simplex. It describes where and how completion evidence was produced.

We use two runtime summaries:

$$\text{support}(t) = \frac{|\{s : v_t(s) > 0\}|}{|\Omega|},$$

$$G_{\text{exp}}(t) = \text{Gini}(\{v_t(s) : s \in \Omega\}).$$

These summaries are operational diagnostics, not universal laws. They summarize whether the evidence condition is localized or broad enough to make a completion certificate eligible for consideration.

Proposition 1 (Local no-new evidence is not a global certificate). *Let $U \subset \Omega$ be a strict subset of the intended source-route space. Suppose all evidence used to support a stop claim C_t is produced inside U , and suppose the workflow observes no-new evidence only for actions inside U . Then the no-new evidence can support local exhaustion over U , but it cannot by itself support global completion over Ω .*

Proof sketch. No-new evidence is conditioned on the source-route actions that were executed. If no events are produced in $\Omega \setminus U$, the workflow has not observed the absence of residual evidence in those unexposed strata. A world in which U is exhausted and $\Omega \setminus U$ contains residual evidence is observationally indistinguishable from a globally complete world under the workflow’s local evidence log. Therefore local no-new evidence rules out additional findings only under the searched condition U ; without additional assumptions about unsearched strata, it cannot rule out residual evidence in $\Omega \setminus U$.

Source-only geometry is too coarse: auditing a file for time-out behavior does not certify that TLS, retry, exception, or cleanup behavior has been audited in that file. Source-route-action geometry is more detailed, but it is sensitive to logging conventions and current experiments do not show a stable advantage over source-route. Source-route is the default because it captures both where and how evidence was produced while remaining interpretable and runtime-computable. The geometry is thus a certificate diagnostic rather than a semantic theory of all possible evidence. It asks whether the current log covers the intended source-route space broadly enough for the proposed stop claim; it does not assert that all routes are equally important or that support/Gini thresholds transfer unchanged across domains.

5 Method

The method has four modules: source-route exposure estimation, eligibility gating, residual-evidence repair, and the final SAFE/CONTINUE/ABSTAIN decision rule. The point is not to prove universal completion; it is to make the runtime evidence condition explicit before a stop claim is accepted.

5.1 Source-Route Exposure Estimation

Input: a runtime evidence log E_t , intended source set S , intended route set R , and their product space $\Omega = S \times R$. Output: exposure counts $v_t(s)$, normalized exposure $p_{\text{exp}}(t, s)$, source-route support $\text{support}(t)$, and exposure concentration $G_{\text{exp}}(t)$. Role: summarize where and how evidence was gathered, so localized stop evidence can be distinguished from broad evidence. Algorithm 1 shows how the four modules are composed at a proposed stop.

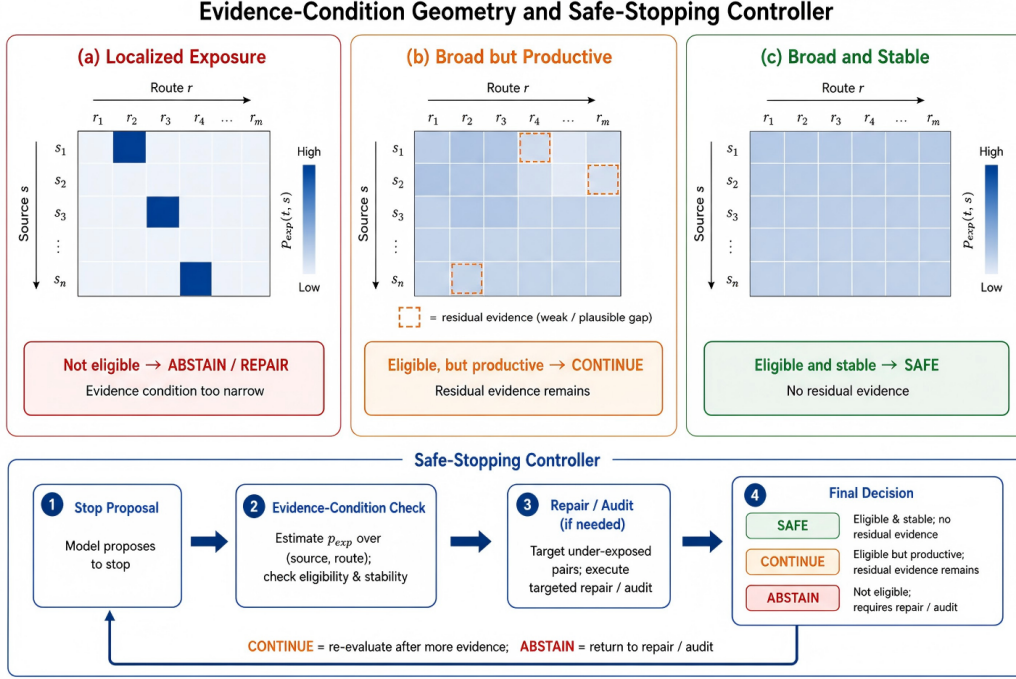


Figure 2: Evidence-condition controller. The controller maps a runtime evidence log to an exposure distribution over the intended source-route space. The support/Gini check is an eligibility test, not a completion proof. If the condition is localized or weak plausible gaps remain, the controller repairs or continues. It returns SAFE only when the evidence condition is broad and repair or audit reveals no residual evidence.

The source-route exposure summaries are operational diagnostics, not universal laws. They summarize whether the evidence condition is localized or broad enough to make a completion certificate eligible for consideration.

5.2 Eligibility Gate

Input: $v_t(s)$, $p_{exp}(t, s)$, $\text{support}(t)$, $G_{exp}(t)$, thresholds τ_s and τ_g . Output: eligible / ineligible. Role: decide whether the current evidence condition is broad enough for a stop claim to be considered.

The eligibility gate is not a completion proof. It is only a test of whether the evidence condition is broad enough for a global stop claim to be considered. In the current implementation, support and Gini thresholds instantiate this operational gate; they are not claimed as universal laws.

5.3 Residual-Evidence Repair

Input: current evidence log E_t , evidence condition Ω , claim C_t , budget B , and repair rule $\mathcal{R}$. Output: repaired log or continued audit targets. Role: probe weak plausible strata for residual evidence when eligibility alone is insufficient.

The repair instance used in the experiments is

$$\text{priority}(s) = \text{under_exposure}(s) \cdot \text{runtime_potential}(s).$$

The under-exposure term gives priority to weak parts of the current completion certificate. The runtime-potential term uses

visible information such as source text, route names, source size, route match counts, or non-oracle ledger signals. It is not allowed to use oracle labels, oracle totals, oracle missing mass, undiscovered true item counts, post-hoc recall, or scorer-visible target distributions. Residual-potential is a mechanism-aligned repair instance, not an optimal active-search algorithm.

5.4 SAFE/CONTINUE/ABSTAIN Decision Rule

Input: eligibility result and residual-evidence repair outcome. Output: SAFE, CONTINUE, or ABSTAIN. Role: accept a stop only when the evidence condition is broad and repair or audit reveals no residual evidence.

The final rule is intentionally conservative. Broad exposure gives completion eligibility, not automatic safety. A SAFE decision additionally requires that repair or audit reveal no residual evidence. If residual evidence appears, the controller outputs CONTINUE; if budget is exhausted without a defensible certificate, it outputs ABSTAIN.

6 Experiments

The experiments are not a discovery benchmark. They are controlled completion-audit tests designed to check whether stop evidence is sufficient to support a completion claim under a declared source-route scope.

We organize experiments around the method design points. First, do naive stop signals create false certification? Second,

Algorithm 1 Evidence-Condition Controller

Require: $\log E_t$, space Ω , claim C_t , budget B , thresholds τ_s, τ_g , repair rule $\mathcal{R}$

Ensure: decision in $\{\text{SAFE}, \text{CONTINUE}, \text{ABSTAIN}\}$

```
1: Compute  $v_t(s)$ ,  $p_{\text{exp}}(t, s)$ ,  $\text{support}(t)$ , and  $G_{\text{exp}}(t)$ 
2:  $\text{eligible} \leftarrow [\text{support}(t) \geq \tau_s \wedge G_{\text{exp}}(t) \leq \tau_g]$ 
3: if not  $\text{eligible}$  and  $B = 0$  then
4:   return ABSTAIN
5: end if
6: if not  $\text{eligible}$  or weak plausible gaps remain then
7:   Audit  $Q = \mathcal{R}(E_t, \Omega, C_t)$  within  $B$ 
8:   if residual evidence appears then
9:     return CONTINUE
10:  end if
11:  Recompute  $\text{eligible}$ 
12: end if
13: if  $\text{eligible}$  and no residual evidence appeared then
14:   return SAFE
15: end if
16: return ABSTAIN
```

is source-route geometry necessary relative to source-only exposure? Third, is eligibility alone sufficient? Fourth, does the full controller avoid unsafe stops without becoming a never-stop rule? Fifth, how does residual-potential repair compare with random and high-potential repair?

Tasks. `policy_docset_v1` is a generated policy document set with four sources covering access control, data handling, release process, and audit exceptions. Its routes are obligation, exception, deadline, and prohibition; the hidden oracle contains 24 policy clauses. `code_repo_v1` is a generated bounded code repository with authentication, payment, storage, and API-client sources. Its routes are compatibility, security, and resilience; the hidden oracle contains 20 code-risk items.

`requests` is an external real-repository audit over six files from a local installed snapshot of the Python package: adapters, API, authentication, models, sessions, and utilities. Its routes are TLS, timeout, exception, and compatibility. `urllib3` is an external real-repository audit over connection, connection-pool, pool-manager, response, retry, and timeout sources. Its routes are timeout, retry, TLS, exception, and cleanup, giving 30 intended source-route strata. External oracles are pattern-defined from frozen snapshots: stronger than generated toys, weaker than human-annotated completion-audit benchmarks.

Workflow conditions. Homogeneous route reuse assigns multiple agents to the same route across sources. The reused routes are obligation for `policy_docset_v1`, compatibility for `code_repo_v1`, and timeout for the two external repository audits. Route-partitioned audit assigns agents to different routes over the same source set. `urllib3` additionally includes an `extended_audit` condition covering all five routes.

Compared policies and ablations. We compare six decision policies on fixed bounded-audit states. Naive stop-signal acceptance accepts a workflow’s local stop signal as SAFE without conditioning on exposure. The source-only controller computes support over sources but ignores routes; it is the ablation that asks whether touching each file or document is enough. Source-route eligibility-only applies the support/Gini gate and accepts eligible states without residual-evidence repair. Random repair uses the same decision rule as the full controller but chooses repair strata uniformly. High-potential repair chooses strata with the largest runtime potential, ignoring under-exposure. Ours uses source-route exposure, the eligibility gate, residual-potential repair, and the final SAFE/CONTINUE/ABSTAIN rule in Algorithm 1. All six policies are compared on the same fixed trajectories and proposed stop states; differences come from the decision rule and repair target selection, not from different discovery trajectories. Oracle labels and recall are used only for post-hoc scoring.

Seeds, budgets, thresholds, and cost. Generated-task first-pass challengers used a target budget of 4 strata; method validation reran generated-task challengers for 200 seeds. `requests` challengers used 200 seeds and a budget of 4 strata. `urllib3` challengers used 200 seeds and a budget of 5 strata. The external controller validation uses $\tau_s = 0.75$ and $\tau_g = 0.70$. A 0.90 recall threshold is used only for bounded-oracle evaluation of false certification and is not visible to the controller. We separate original workflow cost from post-hoc audit cost: repair cost and confirmation-audit cost count only the additional scanned source lines plus extraction events after a stop state is fixed. Thus safe-state validation can have repair gain 0 but positive cost: the extra probes are confirmation audits that verify no additional route-level evidence appears. For each seed, route assignment, repair ordering, and challenger choice are fixed before oracle scoring. This makes the reported false-certification and repair measurements post-hoc evaluations of fixed trajectories rather than inputs to the controller.

Table 1 summarizes the amount of evidence behind the empirical claims. The four task families supply bounded completion-audit cases with withheld or post-hoc oracles. The simulation stress tests are not used as main task results; they check whether the geometry variables behave consistently under larger controlled sweeps and whether the diagnostic is an artifact of a single task family. Leakage-control details are reported in the anonymous supplementary material; runtime decisions use visible logs and source text, not oracle labels, oracle totals, post-hoc recall, or undiscovered true-item counts.

7 Results

Naive stop signals cause false certification. Figure 3 begins with the failure mode the paper is built to prevent. Across all four tasks, homogeneous route reuse produces localized evidence conditions and would cause false certification if accepted as SAFE. The result is not merely an item-recovery

Table 1: Experimental coverage. The main evaluation combines generated audits, external repository audits, seeded repair/controller validation, and simulation stress tests. Pattern-defined external oracles are used only for post-hoc scoring.

Study	Role	Sources	Routes/strata	Oracle items	Runs/seeds	Main measurement
<code>policy_docset_v1</code>	generated audit	4	4/16	24	200/challenger	false certification, repair gain
<code>code_repo_v1</code>	generated audit	4	3/12	20	200/challenger	false certification, repair gain
<code>requests</code>	external repo audit	6	4/24	298	200/challenger	controller validation
<code>urllib3</code>	external repo audit	6	5/30	699	200/challenger	boundary validation
Controlled geometry	simulation stress test	–	variable	synthetic	2160 runs	metric screening
Exposure localization	simulation stress test	–	18/30/42	synthetic	2700 runs	exposure vs. discovery localization

failure; it is a certificate mismatch between local evidence and a global completion claim. The finding is also consistent with controlled sweeps: in the 2160-run geometry stress test, homogeneous and prompt-diverse local conditions have false-stop rates of 1.000, while route-partitioned and extended audits reduce the rate to 0.637 and 0.546, respectively. In the 2700-run exposure-localization stress test, homogeneous and prompt-diverse conditions again remain near complete false-stop failure (0.998 and 1.000). This motivates source-route granularity as the exposure unit rather than accepting unconditioned stop signals.

Source-route geometry is necessary. The middle panel of Figure 3 is now the explicit source-only vs. source-route ablation. In the homogeneous runs, every task touches all source files, so a source-only controller would see full support. Yet the same states have low source-route support, high concentration, and sub-threshold bounded-oracle recall. Source-only exposure therefore cannot detect the route-level mismatch: auditing a file for timeout evidence does not certify TLS, retry, exception, or cleanup routes in that file. A scalar singleton/Chao-style proxy is included only as a negative control in the supplementary tables. It summarizes item-count sparsity but does not encode where evidence was gathered, so it cannot identify the route-level certificate mismatch between a local no-new signal and a source-route completion claim. This is the source-route granularity design point: completion evidence must track both the source and the route under which it was gathered.

Eligibility alone is not proof. Route-partitioned evidence improves completion eligibility in all tasks, but it does not always justify SAFE. The `urllib3` row is the boundary case. Route-partitioned exposure is broad enough to pass the eligibility gate, but bounded-oracle recall is below threshold and repair/audit remains productive. The controller returns CONTINUE, not SAFE. The extended `urllib3` audit reaches support 1.000, recall 1.000, and SAFE. This boundary is important because it prevents a different shortcut: replacing local-stop acceptance with automatic acceptance of broad exposure. The `urllib3` condition has 24 of 30 source-route strata exposed and 584 of 699 oracle items found, but weak plausible gaps remain and additional audit finds residual evidence. In Figure 3, this is the eligibility-only versus eligibility+repair split. This is the eligibility/proof separation design point: broad exposure makes a state eligible for certification, but residual-evidence

checks determine whether it is actually SAFE.

Full controller avoids unsafe stops without never-stopping.

A naive controller that accepts local stop signals would falsely certify all four homogeneous conditions. The evidence-condition controller rejects these stops because the evidence condition is localized. It also avoids the trivial “never stop” solution: it returns SAFE for the two generated tasks and for `requests` under broad complete conditions, and reserves SAFE for the extended `urllib3` audit rather than the broad-but-incomplete route-partitioned condition. Across the two external repository controller validations, all seeded repair/controller trials have zero false-certification rate after the controller is applied: six `requests` challenger policies and five `urllib3` challenger policies, each evaluated with 200 seeds. These trials do not make the controller complete; they show that the implemented decision rule rejects the unsafe local stops while still allowing SAFE when the external audit state is broad and nonproductive. Figure 4 expands this accounting by comparing all six policies on the same fixed stop states and budgets. Seeded repair states are oracle-unsafe and mostly return CONTINUE because repair remains productive. Seeded safe states check the opposite failure mode: starting from oracle-safe complete states, the controller remains SAFE across 6000 order/budget perturbations. Both sides have FCR 0; safe-state coverage is 1. The controller is therefore not a never-stop rule: it rejects unsafe productive states without becoming over-conservative once the source-route evidence condition matches the completion claim. This is the non-never-stop decision-behavior design point.

Repair policy comparison. Figure 4 also summarizes the repair boundary through repair gain and repair cost. Residual-potential provides positive repair evidence but is not proven optimal. On `requests`, residual-potential and high-potential are identical at source-route granularity. On `urllib3`, residual-potential recovers more total new evidence, but high-potential is similar or slightly better in cost-normalized evidence. The correct conclusion is that residual-potential is a mechanism-aligned repair instance that can expose residual evidence, not an optimal active-search method. The supplement reports confidence intervals, task-level repair-gain details, and the underlying seeded safe-state and repair-state counts. This bounded conclusion is reinforced by metric screening. In the 2160-run controlled-geometry study, cover-

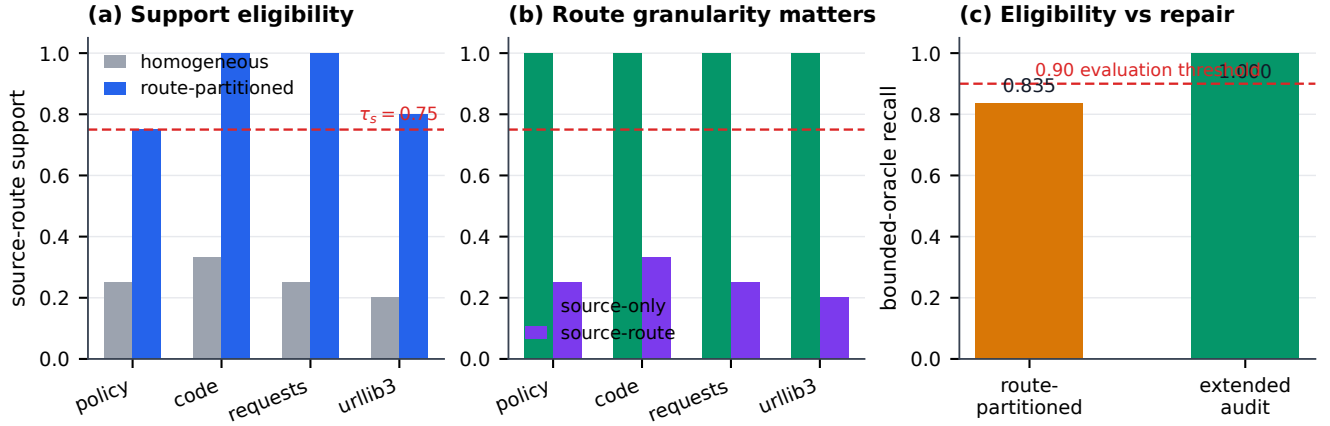


Figure 3: Main diagnostic and ablation. Left: homogeneous base states have low source-route support; broader audits increase support. Middle: source-only vs. source-route ablation. Source-only support is full in homogeneous runs, but source-route support remains localized, exposing the route-level certificate mismatch. Right: `urllib3` boundary as eligibility-only vs. eligibility+repair. Route-partitioned audit is broad but incomplete, while the extended audit reaches completion. Dashed lines mark the operational support and recall thresholds used for evaluation.

age Gini is the strongest single diagnostic (AUROC 0.995), while the more elaborate compound risk quantity is weaker (AUROC 0.831). A leave-world-out check over three synthetic worlds keeps coverage Gini stable (mean AUROC 0.996, minimum 0.994). In the 2700-run exposure-localization study, exposure and discovery Gini are both predictive (AUROC 0.845), which supports the broader localization argument without overclaiming that residual-potential is uniquely optimal. This is the residual-evidence repair design point: repair is a bounded probe for remaining evidence, not a claim of optimal active search.

Sensitivity and audit checks. Sensitivity details are moved to the supplementary material. Threshold and budget sweeps over the external controller validations scan $\tau_s \in \{0.50, 0.60, 0.70, 0.75, 0.80, 0.90\}$, $\tau_g \in \{0.50, 0.60, 0.70, 0.80, 0.90\}$, and repair budgets $\{1, 2, 4, 6, 8\}$. These sweeps mainly test unsafe productive states; changing thresholds and budgets changes the SAFE/CONTINUE/ABSTAIN mixture and repair cost without creating false certifications. The seeded safe-state rows in Figure 4 complement this by testing completed oracle-safe states. Across threshold settings, `requests` has SAFE-rate range 0.000–0.000, CONTINUE-rate range 0.995–1.000, and max FCR 0; `urllib3` has SAFE-rate range 0.000–0.000, CONTINUE-rate range 0.990–1.000, and max FCR 0. Budget sweeps keep max FCR 0 while moving abstention up to 0.365 for `requests` and 0.360 for `urllib3`. Oracle construction, scalar negative controls, route-match consistency checks, workflow-shape/interface validation, leakage controls, seeded controller details, and regex examples are reported in the anonymous supplementary material rather than used as main effectiveness evidence.

8 Discussion and Validity

The controller certifies a relationship between a stop claim and the evidence condition that produced its support. It does not certify that the world contains no further facts, that all future searches would fail, or that source-route is the only possible abstraction. This distinction separates the paper from total-recall and active-search optimization: recall is used only for post-hoc false-certification measurement, while the run-time decision uses exposure, eligibility, and repair outcomes. Multi-agent agreement is likewise not rejected as useless; it is rejected only as an unconditioned completion certificate.

The main internal threat is leakage from the hidden oracle into repair or stop decisions. The protocol constructs oracles offline, freezes trajectories before scoring, and restricts runtime features to visible logs, source-route identifiers, source text, lexical match counts, and non-oracle ledger signals. Thresholds are operational test points, not learned universal constants. Support and Gini measure exposure breadth and concentration rather than semantic adequacy; domains with unequal route importance should declare weights or route priorities before execution.

The experiments cover generated document/code audits and two real-repository audits with pattern-defined oracles. This is broader than a toy-only validation but narrower than a human-annotated benchmark of completion claims. The strongest external claim is therefore limited: local source-route evidence can create false completion certificates, and an evidence-condition controller can prevent this class of unsafe stop in bounded audit settings. The paper does not claim universal open-world completeness or optimal repair search.

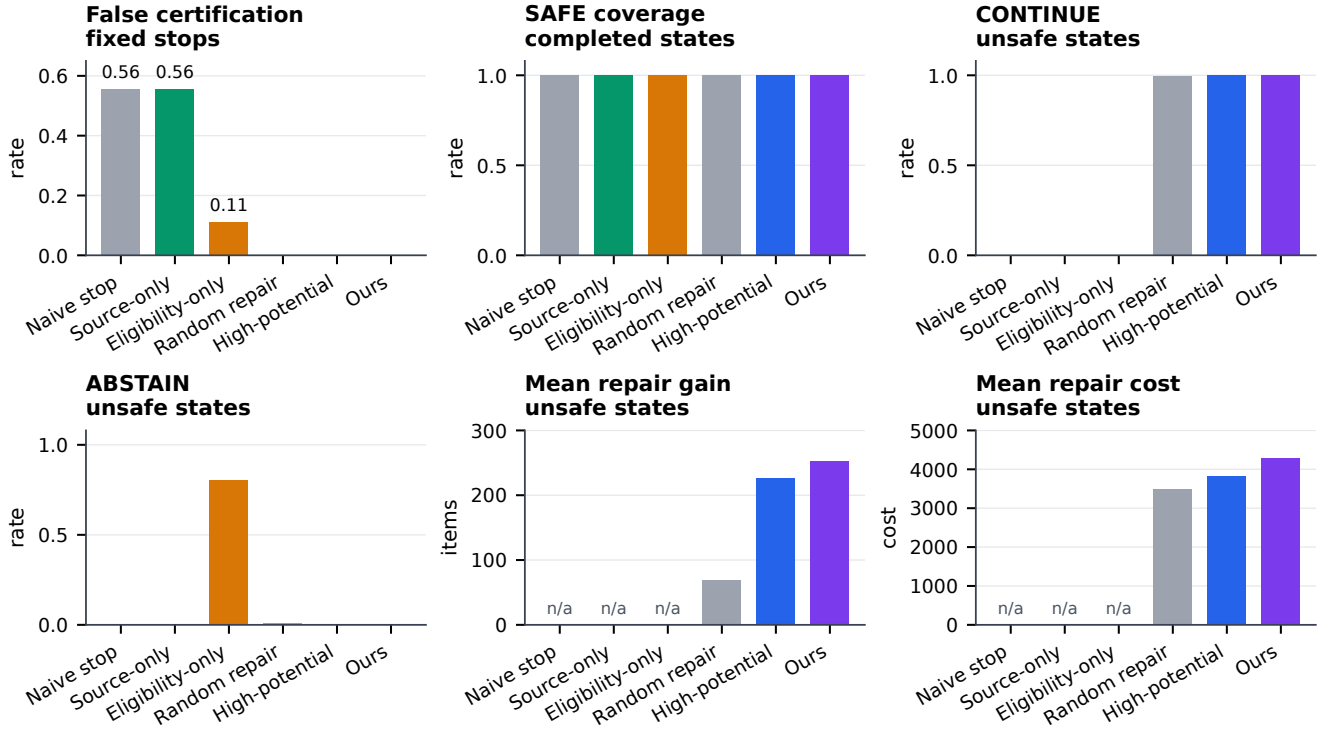


Figure 4: Controller variant comparison on the same fixed stop states, fixed repair budgets, and oracle-only post-hoc scoring. Naive stop and Source-only accept unsafe localized stops; Eligibility-only removes most but not all false certifications because broad exposure is not proof. Repair variants use the same decision rule but different repair target choices. Ours combines source-route exposure, eligibility, residual-evidence repair, and the SAFE/CONTINUE/ABSTAIN rule.

9 Limitations and Supplement

The current evidence supports a diagnostic and controller in bounded completion-audit subclasses. External oracles are pattern-defined rather than human annotated; support/Gini thresholds are operational; source-route strata are task-designed; and residual-potential is not proven optimal. The anonymous supplementary material records reproducibility details that are not main effectiveness evidence: the scalar negative control, external oracle construction summary, workflow-shape/interface pilot, seeded safe-state breakdown, route-match consistency sanity check, leakage-control matrix, and route-wise pattern examples. A future claim-verification audit with support, contradiction, exception-path, configuration-default, and scope-boundary routes would test the same certificate-mismatch principle beyond item-discovery audits.

10 Conclusion

This paper studies completion decisions in bounded completion-audit workflows with declared source-route scopes and unknown residual workload. The central observation is simple but consequential: local evidence does not automatically support a scope-wide completion claim. Stop signals such as no-new findings, agreement, or stable summaries are conditioned on where and how the workflow searched.

Source-route evidence-condition geometry makes this conditioning explicit. The evidence-condition controller uses that geometry to reject unsafe local stops, repair weak plausible strata, and reserve SAFE for cases where the evidence condition is broad and no residual evidence appears. Across generated and external bounded audits, homogeneous route reuse consistently creates false certification risk, while broader source-route exposure improves completion eligibility without being sufficient by itself. The main contribution is not a claim that a particular repair heuristic is optimal. It is a diagnostic and control principle: completion certificates should be judged by the evidence condition that produced them.

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